### Program Start

```
checkmygrade.py (3.14.2)

Python 3.14.2 (v3.14.2:df793163d58, Dec  5 2025, 12:18:06) [Clang 16.0.0 (clang-1600.0.26.6)] on darwin
Enter "help" below or click "Help" above for more information.
>>>
================================================================================================================================ R
PROGRAM STARTED
===== Welcome to CheckMyGrade =====
1. Register
2. Login
Choose option: |
```

This screenshot shows the program starting in the Python environment. The system displays the main menu for the CheckMyGrade application, allowing the user to either register a new account or log in to an existing account.

# User Registration

```
checkmygrade.py (3.14.2)

Python 3.14.2 (v3.14.2:df793163d58, Dec  5 2025, 12:18:06) [Clang 16.0.0 (clang-1600.0.26.6)] on darwin
Enter "help" below or click "Help" above for more information.
>>>
================================================================================================================================ R
PROGRAM STARTED
===== Welcome to CheckMyGrade =====
1. Register
2. Login
Choose option: 1
Email: testuser@sjsu.com
Password: 12345
Role (student/professor/admin): student
User registered
```

This screenshot demonstrates the user registration functionality. The user provides an email, password, and role (student, professor, or admin). The system encrypts the password and stores the credentials in the login.csv file.

### User Login

```
================================================================================================================================
PROGRAM STARTED
===== Welcome to CheckMyGrade =====
1. Register
2. Login
Choose option: 2
Email: testuser@sjsu.com
Password: 12345
Login successful
```

This screenshot shows the login process where the user enters their registered email and password. The system validates the credentials by decrypting the stored password and grants access if the authentication is successful.

## Add student

```
===== CheckMyGrade Menu =====
1. Add Student
2. Display Students
3. Search Student
4. Update Marks
5. Delete Student
6. Sort Students by Marks
7. Average Marks
8. Median Marks
9. Exit
Enter your choice: 1
Email: john@sjsu.com
First Name: john
Last Name: rover
Course ID: 201
Grade: A
Marks: 97
Student added successfully
```

This screenshot demonstrates the functionality for adding a new student record. The system collects student details including email, first name, last name, course ID, grade, and marks, and stores the information in the students.csv file.

## Display students

```
===== CheckMyGrade Menu =====
1. Add Student
2. Display Students
3. Search Student
4. Update Marks
5. Delete Student
6. Sort Students by Marks
7. Average Marks
8. Median Marks
9. Exit
Enter your choice: 2
['email', 'first_name', 'last_name', 'course_id', 'grade', 'marks']
['zoe@sjsu.com', 'zoe', 'doever', '2424', 'A', '98']
['sarasebastian@sjsu.com', 'sara', 'sebastian', '12', 'c', '65']
['john@sjsu.com', 'john', 'rover', '201', 'A', '97']

===== CheckMyGrade Menu =====
1. Add Student
```

This screenshot shows the display functionality where the system reads all student records from the students.csv file and prints them on the screen.

## Search student

```
===== CheckMyGrade Menu =====
1. Add Student
2. Display Students
3. Search Student
4. Update Marks
5. Delete Student
6. Sort Students by Marks
7. Average Marks
8. Median Marks
9. Exit
Enter your choice: 3
Enter email to search: john@sjsu.com
Student Found: ['john@sjsu.com', 'john', 'rover', '201', 'A', '97']
Search time: 0.09853816032409668
```

This screenshot demonstrates the search functionality. The system searches for a student record using the provided email address and displays the matching record. It also measures and prints the execution time required for the search operation.

Update student marks

```
===== CheckMyGrade Menu =====
1. Add Student
2. Display Students
3. Search Student
4. Update Marks
5. Delete Student
6. Sort Students by Marks
7. Average Marks
8. Median Marks
9. Exit
Enter your choice: 4
Enter email: john@sjsu.com
Marks: 95
Marks updated
```

This screenshot shows the update operation where the marks of a student are modified. The system searches for the student using the email address and updates the marks value in the students.csv file.

Delete student

```
===== CheckMyGrade Menu =====
1. Add Student
2. Display Students
3. Search Student
4. Update Marks
5. Delete Student
6. Sort Students by Marks
7. Average Marks
8. Median Marks
9. Exit
Enter your choice: 5
Enter email to delete: john@sjsu.com
Student deleted
```

This screenshot demonstrates the delete functionality. The system removes a student record from the students.csv file based on the email address provided by the user.

Sort students by marks:

```
===== CheckMyGrade Menu =====
1. Add Student
2. Display Students
3. Search Student
4. Update Marks
5. Delete Student
6. Sort Students by Marks
7. Average Marks
8. Median Marks
9. Exit
Enter your choice: 6
['email', 'first_name', 'last_name', 'course_id', 'grade', 'marks']
Students sorted by marks:
['zoe@sjsu.com', 'zoe', 'doever', '2424', 'A', '98']
['sarasebastian@sjsu.com', 'sara', 'sebastian', '12', 'c', '65']
```

This screenshot shows the sorting functionality where the system sorts student records in descending order based on their marks.

Average marks calculation

```
===== CheckMyGrade Menu =====
1. Add Student
2. Display Students
3. Search Student
4. Update Marks
5. Delete Student
6. Sort Students by Marks
7. Average Marks
8. Median Marks
9. Exit
Enter your choice: 7
Average marks: 81.5
```

This screenshot demonstrates the calculation of average marks. The system reads all student marks from the dataset and computes the mean value.

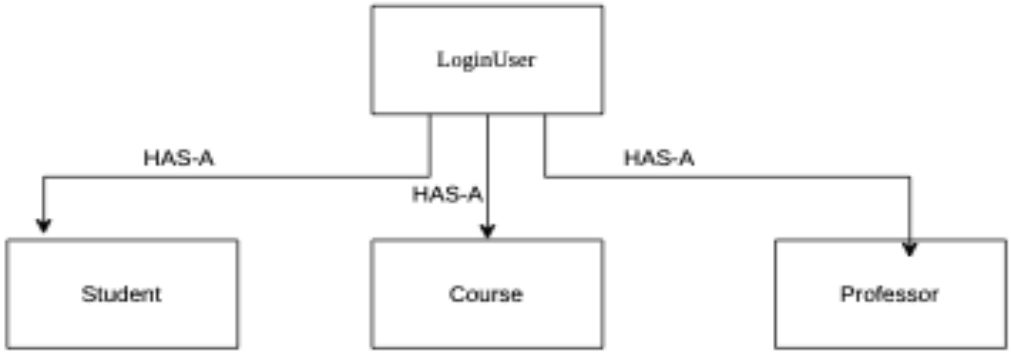
Median marks calculation

```
===== CheckMyGrade Menu =====
1. Add Student
2. Display Students
3. Search Student
4. Update Marks
5. Delete Student
6. Sort Students by Marks
7. Average Marks
8. Median Marks
9. Exit
Enter your choice: 8
Median marks: 81.5
```

This screenshot shows the median marks calculation. The system determines the middle value from the list of student marks using the median function.

## Class Relationship Diagram (IS-A / HAS-A)

This diagram illustrates the relationships between the primary classes in the CheckMyGrade system. The **LoginUser** class handles authentication, while the Student, Course, and Professor classes store system data. Since the system interacts with these entities, the relationship between them is represented as **HAS-A**.


LoginUser
HAS-A
HAS-A
HAS-A
Student
Course
Professor